Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination

June 2025

Max. Marks : 100 Duration : 3 Hrs

Class : TE EXTC/CSE Semester : VI

Course code: OE IT2 Branch : EXTC/CSE

Name of the course : Software Engineering

Instruction:
(1)All questions are compulsory.
(2)Draw neat diagrams and write equations wherever necessary.
(3)Assume suitable data if necessary.

Course Outcomes: At the end of the course student will be able to:
(1) Illustrate various software process models
(2) Analyze software requirements
(3) Design system models with respect to function and object oriented approach.
(4) Apply the basics of software testing methods for the given scenario.

Q No		Max Marks	CO-BL
Q1a)	a) A local airport decides to build an Air Traffic Control(ATC) Assistant system , a high risk project , to manage 20M aircraft in vicinity of airport. The ATC Assistant has several functions. Using the system it is possible to track location of aircraft on ground or in air, and keep track of who is controlling the aircraft. It provides a decision support system for tracking and altering the states of aircraft in designated air traffic zones.There is a module that is used to train air traffic controllers through ATC simulation. A display of weather sensors is also provided. Aircrafts are identified whenever they enter a controlled air space either through a TRACON. Aircraft leaving the control space are assigned to an en route center and removed from the list. Choose the best process model for this system and discuss how the model is applied to the project. Justify why it is better than other models highlighting the problems you face while using other models. b) Differentiate between incremental and iterative model	6+4	1-5
Q1b)	a) Teenagers who run a lawn mowing(cutting) service tend to struggle with administration tasks such as writing quotes and collecting payments. Aruna, who runs a mowing service, currently spends approximately two hours extra each week, besides mowing, on administration. The system that will be developed is expected to save Aruna's time. Additionally, approximately 5% of existing customers are unsatisfied as, for example, their lawn was not mowed as originally arranged.A new system needs to be developed, that supports quotes, cash and card payments, reminders, scheduling, and maps for directions	6+4	3-4

	for a teenager's mowing service. You are to implement the system using database and APIs. Currently there are only few team members available to work on the project. Draw Use Case diagram for the software. b) Identify the fundamental activities common to all process models? **OR** Draw sequence diagram for case study given in Q1B (Teenagers who run a lawn mowing…). Identify the different notations that are part of a sequence diagram.		
Q2a)	Identify best software architecture that a video streaming platform similar to Netflix will use . Identify the different components in the architecture.The application should be scalable and support the following requirements. ▪ Content Creator should be able to upload content at a designated time. ▪ The viewers are able to watch the video on different platforms (TV, mobile-app, etc.). ▪ Users should be able to search videos based on the video titles. ▪ The system should support sub-titles for the videos as well. List 2 differences between software architecture and software design.	2+6+2	3-4
Q2b)	If you are interested in modelling application behaviour that involves several objects collaborating with each other, state chart diagram can be used. Is the given statement true or false? Justify. Draw state chart diagram to represent the states of microwave **OR** How is state and event connected in a state chart diagram and what are the different types of event? Draw state chart diagram for ATM machine.	5+5	3-3

Q3a)	Draw control flow graph for the given graph ?
	Calculate cyclomatic complexity using no. of nodes and edges.
	Can cyclomatic complexity become equal to zero? Justify
	
	(1) { int i, j, k;
	(2) for (i=0 ; i<=N ; i++)
	(3) p[i] = 1;
	(4) for (i=2 ; i<=N ; i++)
	(5) {
	(6) k = p[i]; j=1;
	(7) while (a[p[j-1]] > a[k] {
	(8) p[j] = p[j-1];
	(9) j--;
	}
	(10) p[j]=k;
	(11) }
		4+4+2	4-3

Q3b)	Why is it important to include boundary values in your black-box test data? Illustrate with examples in which a test suite developed using black box techniques might give the impression that 'everything is OK", while a test suite developed with whit box testing techniques (for example, branch coverage) might uncover a fault.	10	4-3
Q4a)	List 5 problems of requirement elicitation? How does an SRS document contribute to the success of a software development project? Complete the following sections for online Library Management System. a) User classes and characteristics b) Constraints c) Functional Requirements d) Non-functional requirements	5+1+4	2-4
Q4b)	Arrange the following type of cohesion from best to worst. a)Logical cohesion b)Sequential cohesion c)Communication cohesion d) Temporal cohesion e)Procedural cohesion Illustrate with examples for each.	2+8	3-3
Q5a)	List rule violations that are not acceptable in a DFD. Draw level 0, 1,2 DFD diagram for an employee hiring system in a company. List the features of the system.	2+8	3-4
Q5b)	i)Which design method is often associated with "bottom-up design" in Structured Design? a) Data Flow Diagram (DFD). b) Entity-Relationship Diagram (ERD). c) Structured Analysis. d) Object-Oriented Programming (OOP). ii)Which design concept in Structured Design promotes the use of well-defined interfaces between modules? a) Encapsulation. b) Abstraction. c) Modularity. d) Inheritance. iii)Why is maximising cohesion important in software design? a)It simplifies the code and makes it more readable. b)It reduces the likelihood of bugs and makes the code maintainable. c)It increases the amount of code that can be reused d)It allows better distribution of code across multiple files iv)Which type of UML diagram is used to represent the dynamic behaviour of a system by showing the interactions between objects or components? a)Class diagram b)Sequence diagram c)Use Case diagram d)State chart diagram v)The most desirable form of coupling is a)Control coupling b)Data coupling c)Common coupling d)Content coupling	10	3-3

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